



Dilated spatial–temporal convolutional auto-encoders for human fall detection in surveillance videos

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Abstract

Although methods based on supervised learning have demonstrated remarkable performance on fall detection, these existing fall detection algorithms require a substantial quantity of manually labeled training data. In this paper, we combine dilated convolution and LSTM based on auto-encoder, which can be trained on unlabeled data, further saving time and resources, and a novel fall score is computed based on the high-quality reconstructed frame to detect falls. Extensive experimental results indicate that the proposed method further boosts the performance, achieving recognition rate of 97.1%, sensitivity rate of 93.9% and precision rate of 95.1% on the UR dataset.

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Keywords: Dilated convolution; Auto-encoder; Fall detection; LSTM

1. Introduction

Aging population has become a global issue that all governments have to deal with. According to the reports of WHO, there are 727 million people aged 65 in 2020, and the number will reach over 1.5 billion by 2050 [1]. As the aging population grows, more elderly will need to be cared for, and many safety issues of the elderly will be highlighted. Among various issues that influence the safety of the elderly, falls become one of the major causes leading to elderly casualties [2]. Falls often happen unexpectedly and usually consume shorter time from start to end. For the elderly, failing to timely recognize accidents will result in physical harm and significant psychological impairment. Especially, more and more elderly begin to live alone, which will aggravate the serious consequences of falls. As a result, designing a real-time fall detection system to timely detect and rescue falling elderly becomes more and more critical.

Recently, several researchers have devoted efforts to develop an intelligent monitoring system to detect falls. In general, in light of the involved devices, existing fall detection systems can be categorized as wearable device-based systems, ambient device-based systems and camera device-based systems [3]. The typical methods based on wearable devices mainly depend on embedded sensors, such as accelerometer to effectively recognize the movement and position of object. However, these wearable devices usually have to be placed in a particular location for a long amount of time, which will be uncomfortable for the elderly. Different from wearable devices, methods based on ambient devices primarily collect audio and visual information for detecting fall instances, which does not restrict human activity. Yet, it is laborious to install these sensors throughout the whole surface. Compared to wearable devices and ambient devices, camera devices can provide comprehensive information about posture and position, and have low cost and non-invasiveness. Consequently, these methods based on camera devices, have recently gotten a lot of attention.

The fall detection systems based on camera devices can be divided into methods based on hand-crafted features and methods based on deep features. Traditionally, methods based on hand-crafted features mainly rely on body shapes, human postures, and tracking head motions. In [4], frame differencing

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is utilized to obtain the human silhouette for describing falls. Additionally, optical flow can also be adopted to estimate the human posture [5]. However, these hand-crafted features are prone to image noise, brightness change, and occlusion, which will influence performance. In this decade, deep neural networks have already generated a rapid and revolutionary development. Recently, deep learning, particularly convolutional neural networks (CNN) [6], now has a great deal of success in computer vision, such as action recognition and object classification etc. Deep learning algorithms have been adopt to automatically learn efficient deep features of objects for recognizing falls, and have demonstrated remarkable performance on fall detection. However, all existing fall detection algorithms based on deep neural networks are usually trained in supervised manner, which is consuming time to acquire and manually annotate.

To solve the above problem, we propose a novel unsupervised method based on the dilated convolutional auto-encoder and LSTM (DCLSTMAE) for recognizing fall events in video surveillance, aiming to train on unlabeled data. Specifically, we firstly adopts depth colorization to fill holes in the depth image. Then, dilated convolution is utilized to expand receptive field with fewer parameters, and LSTM is incorporated to allow the model to take advantage of temporal dependency. Finally, according to the reconstruction error of DCLSTMAE, fall score can be computed for determining whether or not falls occur. We summarize the main contributions of this paper as follows:

- We establish a new end-to-end network DCLSTMAE for fall detection, which can obtain better spatial–temporal features and train on unlabeled data for saving amount of time and resources.
- We propose to utilize hole filling method to enhance the depth and details on the UR depth images, which is beneficial for obtaining high-quality reconstructed frames and can effectively protect personal privacy.
- We design a novel fall score, derived from the DCLSTMAE's reconstruction error and find out its relationship with normal and fall events in a video, which is effective to detect fall events.

2. Proposed method

In this work, a novel fall detection method based on dilated convolution is proposed. As depicted in Fig. 1, the structure of the DCLSTMAE is mainly comprised of a dilated convolutional encoder, dilated convolutional LSTM (DCLSTM) and a decoder. To begin, continuous video frames are fed into the DCLSTMAE, and dilated convolutional encoder and DCLSTM can be combined to extract spatial–temporal features with behavioral information. Secondly, the decoder in DCLSTMAE is used to reconstruct continuous original frames. After optimizing the loss function in training stage, the optimal DCLSTMAE model can be obtained. Finally, fall score based on reconstruction error can be computed to discriminate fall events in the testing stage.

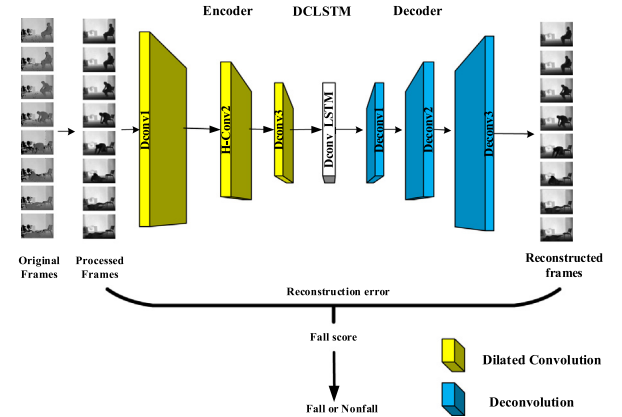


Fig. 1. Framework of the DCLSTMAE proposed in this work.

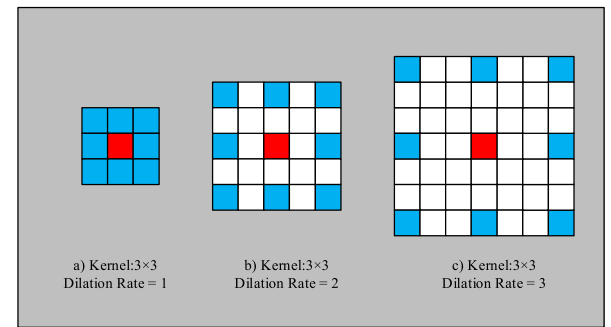


Fig. 2. The dilated convolution framework.

2.1. Dilated convolution

Generally, local evidence is essential when distinguishing aspects such as faces and hands. Yet, in order to judge fall behavior, a thorough understanding of the entire activity is necessary. As a result, dilated convolution is used to increase the receptive field and collect a large number of spatial–temporal data for accurate recognition. By producing zeros between pixels in convolutional kernels, dilated convolution can improve visual resolution, which can be applied to global aspects of human activity without sacrificing resolution.

The dilated convolution is as depicted in Fig. 2. The dilated convolution represents a standard convolutional kernel of size 3×3 when the dilation rate is set to 1. As the dilation rate is greater than 2, a hole (a zero-weighted point) is placed around each point in the matrix to expand the receptive field. The receptive field in (a) (b) and (c) are respectively 3×3 , 5×5 and 7×7 . As the number of inserted holes rises, the size of the receptive field is expanded, whereas the number of parameters in (a), (b), and (c) remain constant. Accordingly, dilated convolution kernel can be utilized to extract more information without increasing the calculation time.

2.2. Dilated convolutional LSTM auto-encoder

In DCLSTMAE, dilated convolution combined with LSTM is very useful for efficient representation of fall behaviors,

more abundant spatial–temporal features can be obtained with fewer parameters.

Encoder: The encoder in DCLSTMAE mainly consists of dilated convolution and pooling. The convolutional encoder of DCLSTMAE uses dilated convolution to extract features from the input frames. In particular, the input of DCLSTMAE is T consecutive frames $X_1, X_2, \dots, X_T$. The information in a filter can be abstracted into a scalar using dilated convolutional layers to parameterize the number of maps, maps size and kernel size. For each dilated convolutional input $X_1, X_2, \dots, X_T$, the hidden feature g_t of the dilated convolutional layer can be represented as:

$$g_t = \sigma(W * X_t + b_t) \quad (1)$$

in which W indicates the weight matrix that can expand the receptive field, σ is the activation function, b_t is the bias, and the symbol $*$ is the dilated convolution.

The feature cube, which is made up of hidden features produced by the dilated convolutional layers, can be down-sampled by the pooling layer. Pooling operations, such as calculating the maximum value among non-overlapping rectangular sub areas, can reduce the spatial dimension of the representation and the number of parameters. Thus, feature size and parameters in the fully connected layer can be efficiently decreased.

DCLSTM: DCLSTM that consists of dilated convolutional layers and LSTM is suitable for obtaining the temporal features of human falls. Rather than employing a sliding window to extract spatial features in CNN, the DCLSTM accomplishes data transmission and processing with three specific gate mechanisms, allowing spatial–temporal information to be employed more efficiently. DCLSTM is shown in Fig. 3, input gate, output gate, and forget gate are three primary gate components used to perform information protection and control. The DCLSTM calculation formulae for these three gate architectures can be obtained as follows:

$$i_t = \sigma(W_{gi} * g_t + W_{hi} * H_{t-1} + W_{ci} \circ C_{t-1} + b_i) \quad (2)$$

$$f_t = \sigma(W_{gf} * g_t + W_{hf} * H_{t-1} + W_{cf} \circ C_{t-1} + b_f) \quad (3)$$

$$\tilde{C}_t = \tanh(W_{gc} * g_t + W_{hc} * H_{t-1} + b_c) \quad (4)$$

$$C_t = f_t \circ C_{t-1} + i_t \circ \tilde{C}_t \quad (5)$$

$$o_t = \sigma(W_{go} * g_t + W_{ho} * H_{t-1} + W_{co} \circ C_t + b_o) \quad (6)$$

$$H_t = o_t \circ \tanh(C_t) \quad (7)$$

where i_t , f_t and o_t are respectively the input gate, forget gate and output gate of DCLSTM. g_t , H_{t-1} and C_{t-1} are the input of the current cell, the output and state of the last cell. W_{gi} , W_{gf} and W_{go} ($\cdot$ represents g , h and c) are respectively the weight matrixes of the input gate, the forget gate and the output gate. b_i , b_f and b_o are respectively the bias of the input gate, the forget gate and the output gate. σ and $*$ are

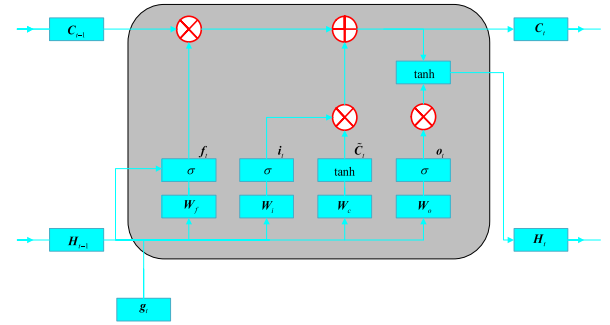


Fig. 3. Illustration of DCLSTM.

respectively the dilated convolution and the sigmoid nonlinear activation function, and $\circ$ denotes the Hadamard product.

Three gate mechanisms are expanded from one-dimensional to multidimensional convolution operations, allowing features to simulate long-range relationships in the temporal domain of video while also capturing the spatial context information of the original data. Thus, DCLSTM can be utilized to extract temporal features for human falls.

Decoder: the function of the decoder is to reconstruct input frames from the spatial–temporal features obtained by the DCLSTM. In general, deconvolution is adopted to reconstruct input frames. The deconvolutional layer in the DCLSTMAE reverses the convolutional process, which can be expressed as:

$$y_t = \sigma(\tilde{W} * H_t + \tilde{b}_t) \quad (8)$$

where y_t is the output of deconvolutional layer, and $\tilde{W}$ and $\tilde{b}_t$ are the weight matrix and bias learnt in the deconvolutional layers, respectively.

2.3. Fall score

Fall events can be recognized by fall score based on reconstruction error of DCLSTMAE. In train stage, normal activity data are only available and fall data are not present. The optimal DCLSTMAE is derived by reducing the reconstruction error on normal activity data. The mean square error can be used to calculate the reconstruction error of DCLSTMAE:

$$J = \min \sum_{i=1}^n \|X_i - Y_i\|_2^2 \quad (9)$$

in which n is the total number of training samples in a batch, X_i and Y_i are the original frames and the reconstructed frames of the DCLSTMAE, respectively.

After the training phase, the trained optimal model of DCLSTMAE can be obtained. In testing phase, the optimal DCLSTMAE that contains historic information can be utilized for distinguishing falls. For normal activity data, the historic information can be helpful to reconstruct the input frames, which can result in a low reconstruction error. However, for the anomaly fall data, the historic information cannot assist in recovering the original frames with the change of appearance or motion. Consequently, the fall score can be computed based

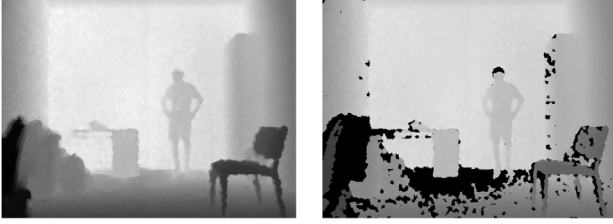


Fig. 4. Processed depth frame and corresponding original depth frame of UR dataset.

on frame reconstruction error. The reconstruction error can be normalized to the range $[0, 1]$ [7], and the fall score for each frame can be calculated as follows:

$$Score(t) = 1 - \frac{R(t) - \min_{k:1 \dots T} R(k)}{\max_{k:1 \dots T} R(k) - \min_{k:1 \dots T} R(k)} \quad (10)$$

in which $R(t)$ is the reconstruction error of test frame and T is the length of a test fall video. The lower $Score(t)$ is that the frame more likely corresponds to a fall event.

3. Experiments

All experiments are implemented in Python using Keras on a PC with GPU GTX-2080Ti, 11 GB RAM. On public fall detection datasets, such as the UR dataset [8] and the Thermal dataset [9], we empirically evaluate our proposed method. In this section, the UR dataset and Thermal dataset are described first. Then, to explore the contributions of different components, related experiments and analyses are concluded. In addition, the performance of the proposed method is compared with the several advanced algorithms.

3.1. Dataset used

UR Dataset – The UR dataset consists of 70 depth videos and 70 RGB videos collected by Microsoft Kinect cameras at 30 frames per second, as well as corresponding acceleration data collected by accelerometers. To protect personal privacy, 70 depth videos provided in a 640×480 resolution, including 40 videos of activities of daily living (ADL) and 30 videos containing falls, are utilized. ADL videos mainly include sitting down, walking and lying down in bed, etc. Falls videos mainly include two fall behaviors that falling from sitting on the chair and from standing position. Nevertheless, due to occlusion or scattering, Microsoft Kinect depth sensors may contain error, which will result in many regions of missing pixels. For learning effective spatial-temporal features, these regions of missing pixels are detrimental. As a consequence, hole filling method is adopted to correct these regions in UR dataset. Corresponding processed depth frame and original depth frame are depicted in Fig. 4.

Thermal Dataset – The Thermal dataset consists of 9 ADL videos and 35 videos containing falls obtained by an Android phone with 640×480 resolution. These videos have a frame rate of either 25 or 15 frames per second. The thermal camera can guarantee individual privacy and identity while still capturing photographs in low-light situations. As shown in Fig. 5, some samples of Thermal dataset are given.

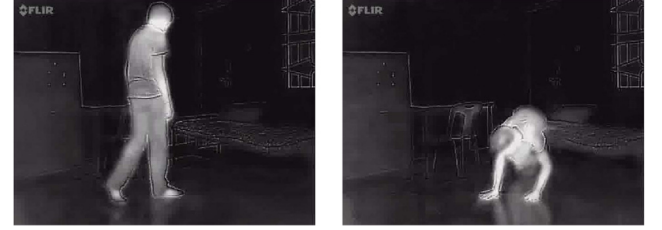


Fig. 5. Samples of Thermal dataset.

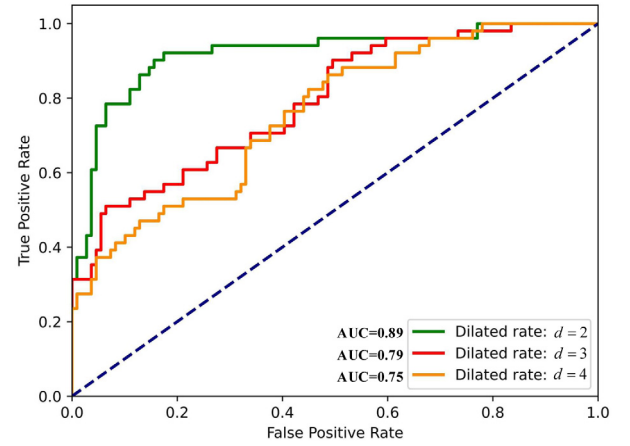


Fig. 6. The ROC curve of different dilated rates in DCLSTMAE.

3.2. Ablation study

In this section, an ablation study on the processed UR dataset is performed. The performance can be evaluated with aspect to the hyper-parameters in our framework, such as dilated rate and the number of the dilated convolutional layers and preprocessed data. After that, the optimal setting of DCLSTMAE can be found by evaluating the variations of the network architecture.

Dilated Rate: To further analyze the performance of the proposed DCLSTMAE, we compare the receiver operating characteristic (ROC) curve of the different dilated rates used in the convolutional layers on processed UR dataset. The ROC curve of the proposed algorithm is shown for the d from 2 to 4 in Fig. 6. From Fig. 6, we note that the optimal performance is at $d = 2$. In addition, when the dilated rate in the proposed method is respectively 2, 3 and 4, the average area under curve (AUC) is respectively 0.89, 0.79 and 0.75. With the increase of the dilated rate, the average AUC of the proposed method is gradually decreased. Furthermore, as shown in Table 1, precision, F1_score and accuracy are gradually decreased with the increase of the dilated rate. This is because larger dilated rate can increase too much holes in convolutional kernel, which will influence the performance. Accordingly, the dilated rate is set as 2 in the proposed method.

The number of layers in the encoder To explore the performance of different encoder structures in the proposed DCLSTMAE, we compare the ROC curve of the different dilated convolutional layers in the encoder. The ROC curve

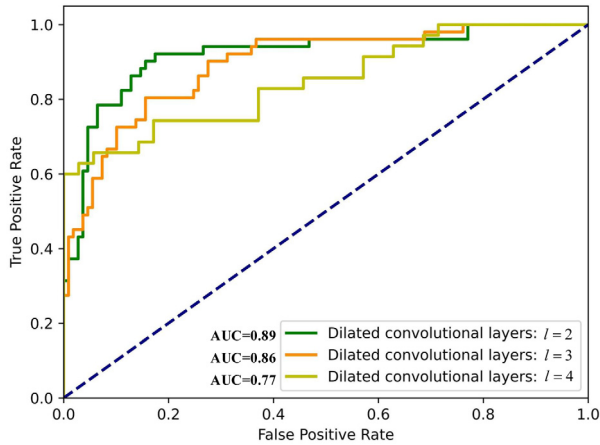


Fig. 7. The ROC curve of different dilated convolutional layers.

Table 1

Metrics of different dilated rates in DCLSTMAE.

Dilated rate	Precision	F1_score	Accuracy
Dilated rate = 2	95.1%	94.7%	97.1%
Dilated rate = 3	89.7%	88.9%	93.4%
Dilated rate = 4	84.1%	85.0%	92.5%

Table 2

Metrics of different convolutional layers in DCLSTMAE.

Convolutional layers	Precision	F1_score	Accuracy
$l = 2$	95.1%	94.7%	97.1%
$l = 3$	90.3%	90.6%	95.3%
$l = 4$	87.3%	87.7%	93.3%

of the proposed algorithm is shown for the d from 2 to 4 in Fig. 7. From Fig. 7, we note that the optimal performance is at $l = 2$. In addition, with the increase of the dilated convolutional layers, the average AUC of the proposed method is gradually decreased. Furthermore, as shown in Table 2, precision, F1_score and accuracy are gradually decreased with the increase of the dilated convolutional layers. It is because larger deeper neural network can result in overfitting when the scale of training data is small. Accordingly, the number of dilated convolutional layers in the proposed method is set as 2.

The Comparison on the UR dataset and processed UR dataset: To further verify the advantage of processed UR dataset, we train and test the proposed method respectively on UR dataset and processed UR dataset. The cross entropy value on processed UR dataset and UR dataset is shown in Fig. 8. From Fig. 8, the cross entropy value on processed UR dataset is significantly lower than that on UR dataset in training phase, which can demonstrate that the proposed method on the processed UR dataset is easier to train. In the test phase, The ROC curve of the proposed algorithm on processed UR dataset and UR dataset is shown in Fig. 9. From Fig. 9, the AUC of the proposed method on processed UR dataset is higher. In addition, the metrics of the proposed method on processed UR dataset and UR dataset are reported in Table 3, we can clearly see that the proposed method based on UR dataset and processed UR dataset has the same model size 12.3 MB, which

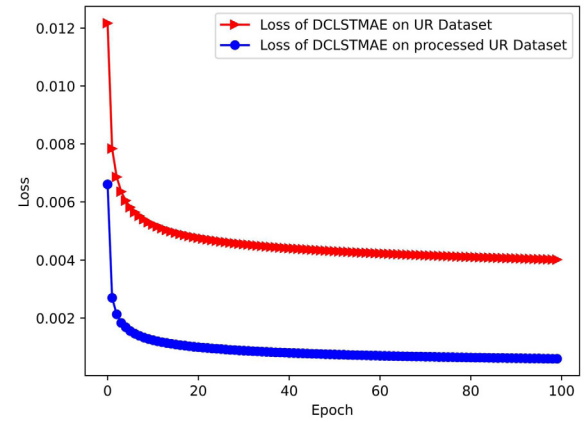


Fig. 8. The cross entropy value of DCLSTMAE on processed UR dataset and UR dataset.

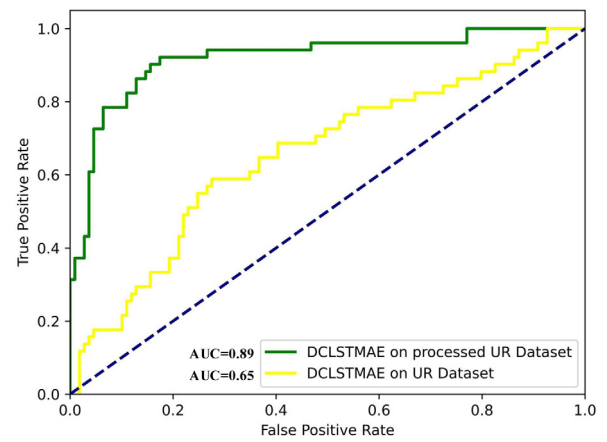


Fig. 9. The ROC curve of DCLSTMAE on processed UR dataset and UR dataset.

Table 3

Metrics of the proposed method on processed UR dataset and UR dataset.

Dataset	Accuracy	Model size	Average time/Epoch
UR dataset	97.1%	12.3MB	128s
Processed UR dataset	86.3%	12.3MB	99s

can verify that the number of parameters is not affected by input. In addition, the accuracy has a great improvement from 86.27% to 97.09%. And it worth noticing that the proposed method based on processed UR dataset is 99 s per epoch, where average processing time is a slightly shorter than that based on the UR dataset. These results effectively show the improvement based on the processed UR dataset.

3.3. Verification on real fall datasets

In this section, we deploy the experiments on the processed UR depth dataset and the Thermal dataset. It is desirable that the trained fall detection model can be directly applied, where the fall score can be computed. The fall score of processed UR depth video is shown in Fig. 10. From Fig. 10, the fall score moves gently in a straight direction in the first 28 frames

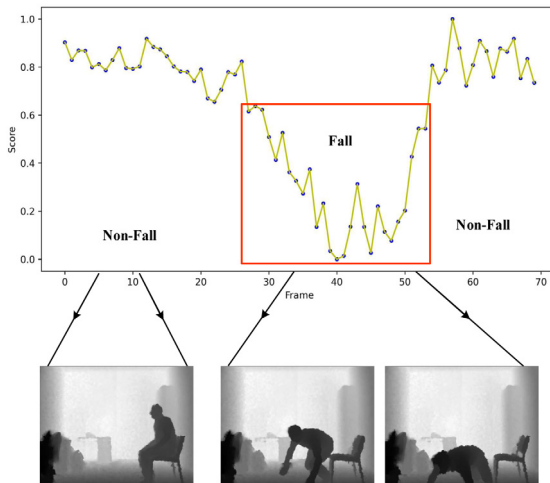


Fig. 10. Score of fall video sample from the processed UR video.

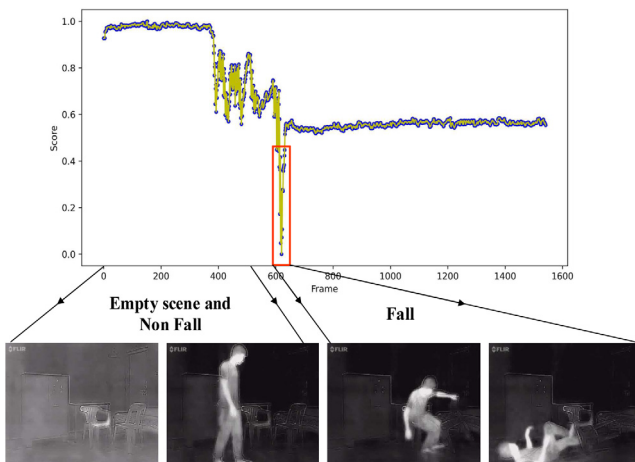


Fig. 11. Score of fall video sample from the Thermal video.

of the processed UR depth video, and the fall score suddenly decreases in the next 25 frames. The fall score and fall video sample can be mapped as shown in Fig. 10, and the fall events are highlighted with red boxes.

The fall score of Thermal video is shown in Fig. 11. From Fig. 11, the fall score represents the empty scene in the first 380 frames and normal human activities from 381th frames to 580th frames of the Thermal video, and the fall score suddenly decreases in the next 20 frames. The fall score and fall video sample can be mapped as shown in Fig. 11, and the fall events are highlighted with red boxes. According to the fall score, a threshold can be set to discriminate falls. For normal activity data, the fall score is higher. Conversely, for the anomaly fall data, fall score will become lower with the change of appearance or motion.

3.4. Comparison with existing methods

To further validate the performance of our proposed method for the fall detection, we firstly conclude the experiments on the UR depth dataset, processed UR depth dataset and the

Table 4

The AUC values compared on different datasets.

Methods	UR dataset	Processed UR dataset	Thermal dataset
CAE	0.38	0.76	0.75
CLSTMAE	0.49	0.82	0.64
The proposed method	0.65	0.89	0.87

Table 5

Comparison with state-of-the-art methods on the UR dataset.

Methods	Sensitivity	F1_score	Precision	Specificity	Accuracy
Area-FD	98%	90%	83%	89.4%	94%
CNN	100%	–	–	92%	95.0%
CNN-LSTM	91.4%	93.1%	94.8%	–	–
CLSTMAE	91.3%	90.5%	89.7%	97.4%	95.8%
The proposed method	93.9%	94.7%	95.1%	98.6%	97.1%

Thermal dataset. The AUC values compared with state-of-the-art methods on different datasets are reported in Table 4. The average AUC of the proposed method is obviously higher than other methods on all the three public datasets. It clearly shows that our method outperforms other existing methods, including CAE, CLSTMAE [10]. Due to process the UR dataset, the average AUC has a great improvement from 0.65 to 0.89. These results adequately prove that the combination of LSTM and the dilated convolution makes significant contribution to fall detection. Furthermore, Table 5 shows experimental results on UR fall dataset, with respect to the evaluation metrics accuracy, specificity, sensitivity, F1_score, precision. As we can see from Table 5, our proposed method can yield detection result with the best accuracy and obviously achieve better performance than traditional method based on hand-crafted features [10] and other state-of-the-art methods based CNN, such as CNN proposed in [11], the CNN with LSTM proposed in [12] and the convolutional auto-encoder with LSTM proposed in [13], which can further verify the effectiveness of the proposed network. This mainly results from the processed data can enhance the depth and details, and also from the end-to-end network DCLSTMAE can obtain better spatial-temporal features.

4. Conclusion

In this paper, we propose an unsupervised DCLSTMAE framework for fall detection, which can save time and resources. To protect personal privacy, the depth image dataset and the thermal data is adopted to train network. Additionally, dilated convolution and LSTM based on convolutional auto-encoder are utilized to automatically extract spatial-temporal features on larger receptive field. Finally, a novel fall score based on the reconstruction error is computed to discriminate falls. In comparison to other existing methods, the experimental results indicate that the DCLSTMAE model can

successfully accomplish accurate detection. In the future, researchers should focus on the approach for dealing with complex surroundings to better protect the elderly.

CRedit authorship contribution statement

Suyuan Li: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – original draft. **Xin Song:** Supervision, Writing – review & editing, Funding acquisition. **Siyang Xu:** Methodology, Writing – review & editing. **Haoyang Qi:** Visualization, Investigation. **Yanbo Xue:** Resources, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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